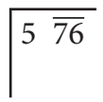
Illustration 1

Find the square root of 576 by long division method.

Step 1:

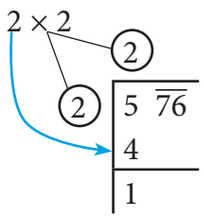
Group the digits in pairs, starting with the digit in the unit's place. Each pair and the remaining digit (if any) is called a period. Put a bar over every pair of digits starting from the right of the given number. If there are odd number of digits, the extreme left digit will be without a bar sign above it.



So, here we have **5 $\overline{76}$**

Step 2:

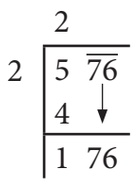
Think of the largest number whose square is equal to or just less than the first period. Take this number as the divisor and also as the quotient. The left extreme number here is 5. The largest number whose square is less than or equal to 5 is 2. This is our divisor and the quotient.



Step 3:

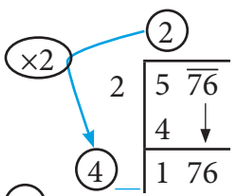
Bring 76 down and write it down to the right of the remainder 1.

Now, the new dividend is 176.



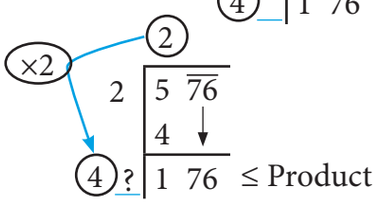
Step 4:

To find the new divisor, multiply the earlier quotient (2) by 2 (always) and write it leaving a blank space next to it.



Step 5:

The new divisor is 4 followed by a digit. We should choose this digit next to 4 such that the new quotient multiplied by the new divisor will be less than or equal to 176.

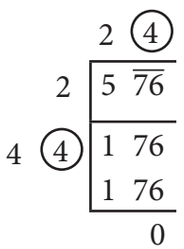


Step 6:

Clearly, the required digit here has to be 4 or 6. (Why?)

When we calculate, $46 \times 6 = 276$ whereas $44 \times 4 = 176$.

Therefore, we put 4 in the blank space and write $44 \times 4 = 176$ below 176 and subtract to get the remainder 0 and the quotient at the top, that is 24 is the square root of 576.



$\therefore \sqrt{576} = 24$.

Illustration 2

In the following example, follow the figures one after another and try to understand what each figure explains, the stage by stage and the gradual computation of computing the square root of 288369.

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